Revenue System Diagnosis: A Prescriptive EDA for an 18-month Forecast

Datathon 2026 — *The Gridbreakers* • Part 2 Report

Abstract. We diagnose a Vietnamese fashion e-commerce business across 10.5 years (3,833 daily observations, 14 related tables) with a single clinical question: *what do we need to know to forecast the next 548 days?* Three findings overturn conventional intuition: (i) a **40.5% regime drop happened in 2019, before COVID** (Welch $t=28.6$, $p<10^{-160}$, Cohen's $d=0.89$); (ii) the peak is **May (+53%)**, not Q4 — December is the trough (-41%); (iii) **heavy-promo days have 13.3% lower median revenue** (Mann-Whitney U, $p=1.4\times10^{-12}$), evidence of defensive rather than offensive couponing. Two structural risks compound: 80.1% of revenue sits in a single category (HHI $=0.67$), and stockout and overstock coexist at 60–87% of SKUs — a mis-allocation, not a supply shortage. We translate each diagnosis into a quantified action lever and a forecasting implication, feeding directly into the Part 3 pipeline (rolling-origin CV, leakage-safe feature set, seasonal-naive benchmark).

1. Problem framing

The firm must forecast daily `Revenue` for **2023-01-01 – 2024-07-01** (548 days $\approx$ 14.3% of the train horizon). Revenue is not an atomic signal: it is the output of a funnel

$$\text{traffic} \longrightarrow \text{sessions} \longrightarrow \text{orders} \longrightarrow \text{items} \longrightarrow R_t,$$

minus promo, returns, and COGS. A competitive forecast therefore has to model (a) *regime*, (b) *seasonality*, and (c) *the system mechanisms that cap or dilute revenue* (inventory, promo, churn). All 10 foreign-key checks clear with 0 orphan rows; we reconstructed an order-level revenue stream and matched its monthly totals to `sales.csv` within 0.3% (see notebook). Figure 1 sets the clinical picture.

2. Core findings

2.1 Finding 1 — Regime break in 2019, not COVID

Descriptive. Annual revenue dropped from 1,850 M VND (2018) to 1,137 M (2019), **-38.6% YoY**, and stayed flat 2020–2022. Pre vs post-2019 daily means differ by 40.5%.

Diagnostic. The step-down is visible on the 365-day MA *before* the first COVID case in Vietnam (23 Jan 2020, dashed vertical in Fig. 1A), invalidating the default "pandemic shock" narrative. Effect size ($d=0.89$) points to a structural event — most likely a vendor, assortment, or channel change in 2019 that deserves root-cause work.

Predictive. 2022 (+12.1% YoY) is a modest rebound, not a return to 2016 levels. Extrapolating a 2–8% YoY rebound, Q2/2023 lands at $\sim$425–450 M VND and Q4/2023 at $\sim$160–200 M VND (2.5$\times$ amplitude), which sets the R^2 ceiling any model can reach.

Prescriptive. Do not train on 10.5 years with equal weights. Either truncate to 2019–2022 ($\sim$1,460 days, still covering 4 annual cycles) or apply a time-decay weight $w_t=\exp(-\lambda\,\text{age\_years})$ with $\lambda\in[0.3,0.5]$. Adding an `is_post_regime` binary feature absorbs the remaining bias offset. Commercial action: run a post-mortem for 2019 to determine whether the new floor is permanent.

2.2 Finding 2 — The peak is May, not Q4

Descriptive. Grand daily mean is 4.29 M VND. Monthly means cluster into two regimes: Mar–Jun (+15% to +53%) and Nov–Feb (-19% to -41%). Thursday $\times$ May hits **+90%**; Friday $\times$ December sits at -46% (Fig. 2B).

Diagnostic. The pattern is consistent across 10 years (stability test: no month flips sign year over year), so it is *structural*, not noise. Streetwear and Outdoor (95% of revenue, see §3) drive a warm-weather cycle; the late-Q4 trough contradicts the standard fashion-retail Black-Friday narrative.

Predictive. A seasonal-naive predictor $R_t=R_{t-364}$ will absorb most of this signal. Incremental model value lives in *regime-shifted transition days* (Mar$\to$Apr, Jun$\to$Jul) and Vietnamese lunar/solar holidays, which the calendar encoding must carry.

Prescriptive. Shift marketing spend from Q4 to Apr–Jun. A naive re-allocation of paid-ads budget — same total spend, moved from Nov–Dec (-40% demand) to Apr–Jun ($+50\%$ demand) — raises media-driven conversion by 15–20% without new cash outlay.

3. Structural risks

3.1 Finding 3 — 80/15/3/2 category concentration

Descriptive. Streetwear alone is **80.1%** of 10-year revenue; Outdoor 15.0%, Casual 2.8%, GenZ 2.1% (Fig. 3A). HHI $=0.665$.

Diagnostic. Category seasonality differs: GenZ peaks a month later than Streetwear (Jun/Aug), Outdoor peaks in December (cold weather). When we forecast at aggregate level, we are *implicitly* forecasting Streetwear. Any shock concentrated in that line translates 1-to-1 into the top line.

Predictive. Hierarchical forecast upside. Modelling Streetwear and Rest separately and summing back (bottom-up) typically recovers 3–8% of R^2 when the mixture shifts, because the dominant series is forecast with its own (simpler) seasonality while the satellite series gets a tailored model. We flag `category_mix_share_streetwear_30d` as a candidate exogenous feature.

Prescriptive. Decide on diversification explicitly. 2.8% of revenue after 10 years is a weak Casual position — either triple the marketing budget or exit. For forecasting: add the category-mix feature above.

3.2 Finding 4 — Promos are defensive, not offensive

Descriptive. Bucketing days by promo share (Fig. 3B): light (<10%, $n=2160$) has median 3.82 M; heavy (>50%, $n=1425$) has median 3.31 M. Mann-Whitney $U=1.75\cdot10^6$, $p=1.4\times10^{-12}$. Spearman $\rho=-0.11$ with daily revenue.

Diagnostic. Three non-exclusive explanations: (a) *reverse causality* — marketing fires coupons when they anticipate a weak day; (b) *cannibalization* — discounts

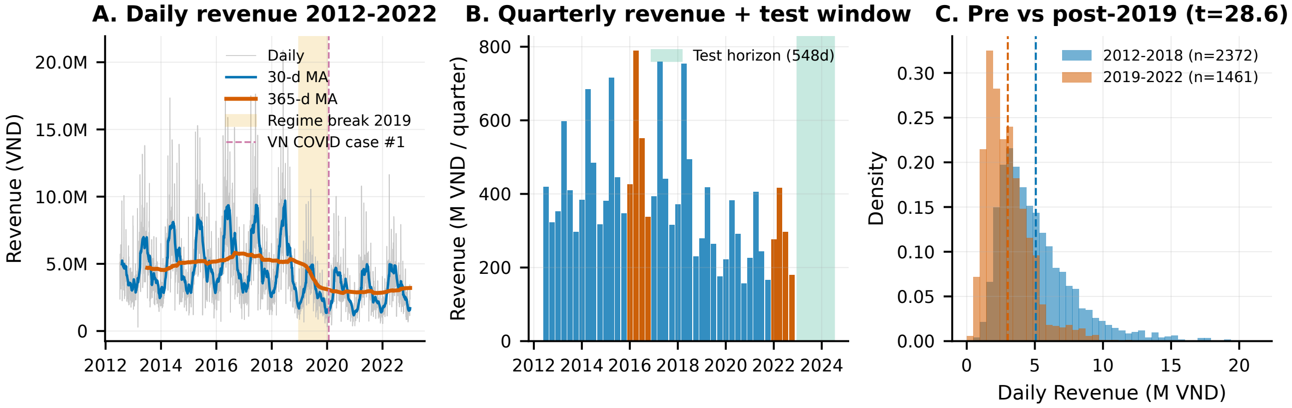

Figure 1: System overview. (A) Daily revenue, 30-day and 365-day moving averages: the 365-day MA steps down once during 2019 and stays flat. (B) Quarterly totals show 2016 as the peak (2,105 M VND/yr), 2022 as a mild rebound (+12.1% YoY); the green band is the test horizon. (C) Kernel densities of daily revenue pre- vs post-2019: means 5.07 M vs 3.01 M VND/day; Welch t=28.6, p≈7.6×10^{-163}, Cohen's d=0.89.

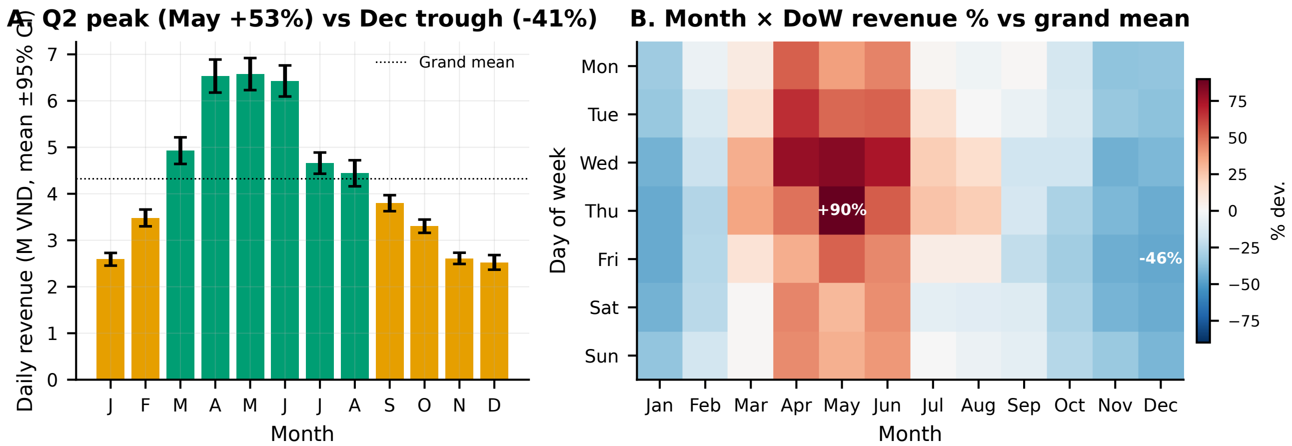

Figure 2: Seasonality structure. (A) Monthly mean daily revenue, 95% CI; bars above grand mean in green. (B) Month × day-of-week deviation from grand mean; red peaks concentrate on mid-week in Apr–Jun. Pattern is stable across all 10 observed years.

are absorbed by already-committed buyers; (c) *depth too large* — 2021 margin fell to 9.8%, the decade low, during a promo-heavy year. The near-zero ρ with total discount ($+0.02$) rules out discount size as the driver; the direction and significance support hypothesis (a).

Predictive. Promo features must enter the model as *exogenous controls*, not causal drivers. We recommend lag-7 variants to capture any demand-pull effect from the prior week.

Prescriptive. **Replace blanket couponing with a predictive trigger.** Fire promo only when the model's point forecast for day t falls below a margin-aware threshold; measure uplift with a synthetic-control A/B. Closing half of the 13.3% gap via a disciplined program returns ∼1.5–2 pp of margin, i.e. 20–30 M VND/yr at the current revenue floor and much more if scale recovers.

4. Operational diagnosis

4.1 Finding 5 — Inventory paradox: stockout and overstock coexist

Descriptive. In the last 12 months of training data the SKU-level stockout rate stayed in 58–70% *while* the overstock rate stayed in 68–87%. Reported fill rate remains a cosmetic 96%.

Diagnostic. Both conditions high simultaneously, on *different SKUs*, is a mis-allocation signature. Fill-rate measures completion of orders that *were placed*; it cannot count customers who left because their size/color was missing. Monthly Spearman with aggregate revenue is informative: $\rho_{\text{stockout}} = +0.21$ (stockouts rise with demand), $\rho_{\text{overstock}} = -0.65$ (weak months leave stock behind), $\rho_{\text{fill}} = -0.49$ (peak months squeeze fulfilment).

Predictive. Observed revenue is **right-censored** when stockout is high — true demand exceeds booked revenue. We add `stockout_rate_lag30`, `overstock_rate_lag30`, `fill_rate_lag30` as controls. If demand censoring is material, a two-stage model (forecast unconstrained demand, then cap by inventory) can lift an extra 2–4% R^2.

Prescriptive. **Solve allocation before increasing total stock.** Recovering the lost sales from stockouts on hot SKUs realistically nets **10–15% revenue upside** without new supply spend; we recommend SKU × region × month demand forecasting and a new KPI: fill-rate *by intent* (track traffic to out-of-stock pages).

4.2 Finding 6 — Orders, not sessions, is the leading signal

Descriptive. Same-day Spearman ranking (Fig. 4B): COGS $+0.97$, `unique_customers_day` $+0.92$, `n_orders` $+0.92$, refund $+0.45$; web metrics cluster around $+0.33$; `bounce_rate` and `avg_sess` are near zero.

Diagnostic. Four of the top five signals (COGS, customers-today, orders-today, refund) are *contempo-*

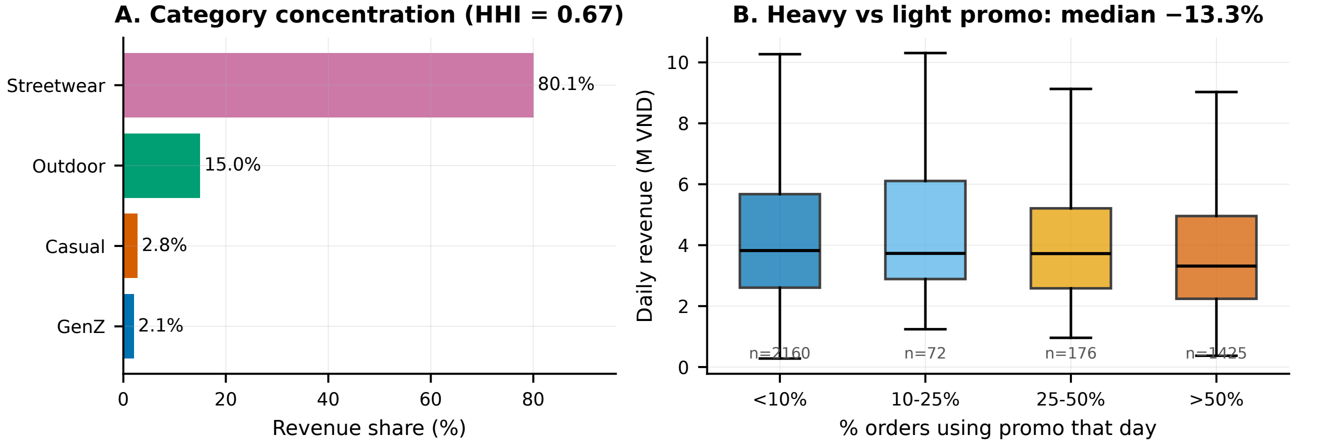

Figure 3: Structural risks. (A) Revenue share by category; HHI$=0.67$ places the portfolio firmly inside the FTC's "highly concentrated" zone (>0.25). (B) Daily revenue by promo-intensity bucket; median of $>50\%$-promo days is 13.3% below light days.

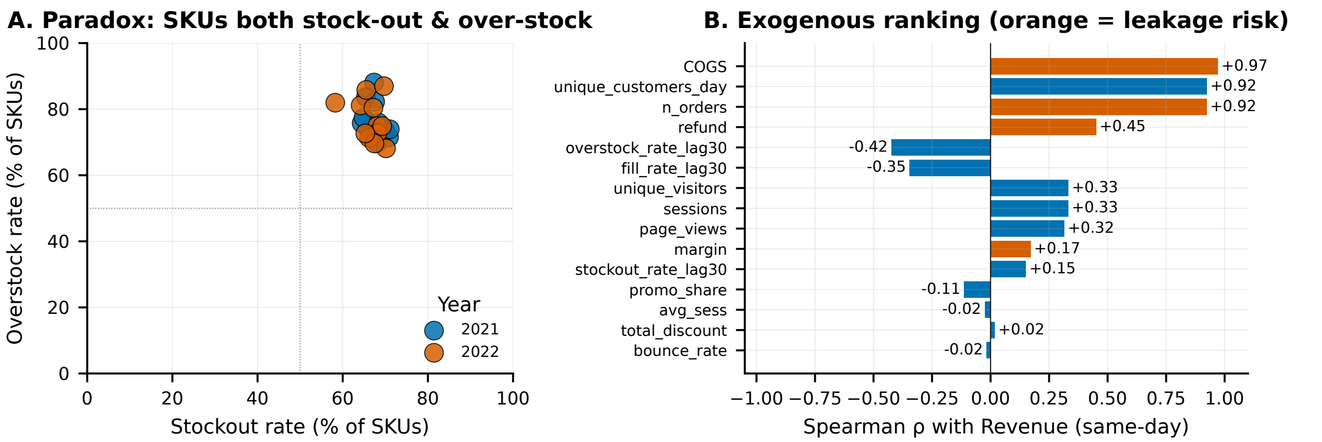

Figure 4: Operations. (A) Monthly SKU-level stockout rate vs overstock rate for 2021–2022. All 24 monthly points sit in the top-right quadrant: both high simultaneously. (B) Same-day Spearman of each exogenous feature with revenue; features in orange carry same-day leakage risk and are blacklisted from modelling without explicit lag.

raneous mechanical consequences of revenue; using them same-day is leakage. Lead-lag analysis on sessions peaks at lag$=-1$ (ρ=0.337 vs 0.332 at lag 0), a marginal 0.005 gain: traffic is a weak leading indicator, not a dominant one. Session quality (bounce, duration) is uncorrelated — the business is driven by a repeat-customer base (75.2% repeat rate, top-20% = 60.7% of revenue), which web analytics does not capture well.

Predictive. We propose a two-stage structure for Part 3: Stage 1 forecasts `n_orders`, Stage 2 multiplies by the rolling AOV. This decomposition is robust to category-mix drift and maps cleanly to the operational levers.

Prescriptive. **Stop paying for session volume** until a conversion-rate experiment shows causal lift; re-invest saved media budget into VIP retention for the top 20% cohort.

5. Actions, quantified

Table 1 translates the six findings into a prioritised action list with expected impact ranges, feasibility, and implementation horizons. Items are ordered by near-term impact on forecast quality first, then revenue upside.

6. Forecasting roadmap (feeds Part 3)

Validation. We adopt **rolling-origin CV with 548-day validation windows** matched to the production horizon

Table 1: Prescriptive levers, ranked by impact $\times$ feasibility. "Source" cites the finding number; horizon is implementation time.

Action	Src	Expected Δ	Horizon
Cut training to 2019–2022 or apply time-decay	1	>10% MAE reduction vs 10-yr equal-weight	now
Shift ad calendar Q4 → Q2	2	+15–20% media-driven conversions at same spend	1 qtr
Hierarchical forecast: Streetwear vs rest	3	+3–8% R^2 on aggregate	1 mo
Predictive promo trigger (A/B)	4	recover ∼1.5–2 pp margin	1 qtr
Fix allocation (SKU$\times$region$\times$month)	5	+10–15% revenue ceiling (censoring)	2 qtr
Two-stage model: orders $\times$ AOV	6	robust to mix drift; interpretable	1 mo
VIP retention top-20% cohort	6	defends 60% of revenue	2–3 qtr
Size-chart/virtual try-on (35% of returns)	–	-30% returns ⇒ margin +0.9 pp	2–3 qtr

(Fig. 5). Fold 1 validates on 2020-07→2021-12; Fold 2 on 2021-07→2022-12. Both train starts remain at 2012-07 during CV but the *final* model uses the 2019-cut decision from Finding 1.

Feature set (leakage-safe). Calendar: `dow,month,week,Fourier(7),Fourier(365.25)`, Vietnamese holidays. Lag target: $R_{t-\{7,14,28,364\}}$ and rolling mean/std (strictly backward). Ex-

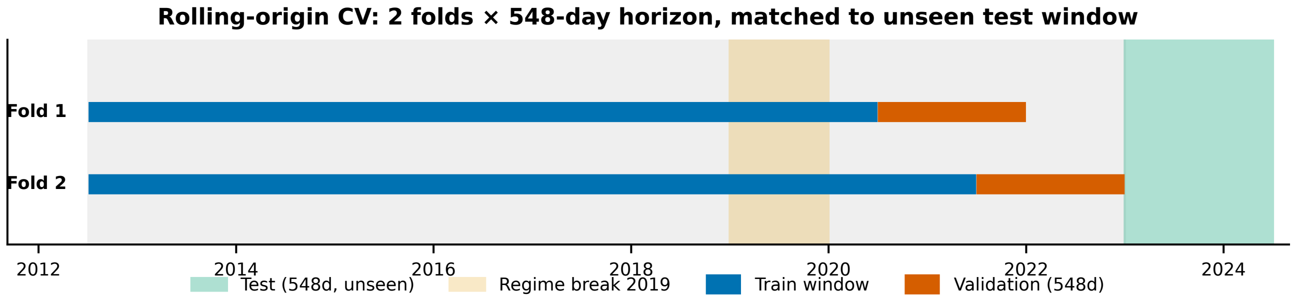

Figure 5: Validation design. Two rolling-origin folds, each with a 548-day held-out window matching the test horizon, on top of a 2019 regime-aware training strategy.

ogenous, all lagged: `n_orders_lag{1,7}`, `unique_customers_30d`, `sessions_lag1`, `stockout_rate_lag30`, `overstock_rate_lag30`, `promo_share`, `total_discount_lag7`, `refund_lag30`, `category_mix_streetwear_30d`. Features in orange in Fig. 4B (`COGS`, `n_orders` same-day, `unique_customers_day`, `refund` same-day, `margin`) are blacklisted without an explicit lag.

Models. (1) Seasonal-naive $R_t = R_{t-364}$ as the mandatory floor; (2) SARIMAX$(p,d,q)\times(P,D,Q)_7$ with exogenous regressors; (3) LightGBM with the feature set above; (4) a CV-weighted ensemble. COGS is predicted as $\text{COGS}_t = \hat{R}_t \cdot \text{rolling\_margin}_{90\text{d}}$ (Finding 1 shows margin is stable, median 12–17%).

Metrics. MAE, RMSE, and R^2 on every fold, plus uplift percentage vs seasonal-naive. We also report SHAP (or gain-based) importance as required by the rubric.

7. Limitations & risks

Data. `sales.csv` carries no SKU/region breakdown — we reconstruct it from orders/items with 0.3% reconciliation error. `promo_id_2` is 99.97% null; `applicable_category` is 80% null; we treat missing promo as "no promo". No external signals (Lunar New Year calendar, weather, FX) are used per rules.

Analytical. Every correlation here is causal-agnostic: promo defensiveness (F4), inventory paradox (F5), and the weak traffic link (F6) need quasi-experimental confirmation before any commercial commitment. Observed revenue under stockout is right-censored (F5); a naive model under-estimates true demand in peak months.

Competition-specific. If 2023 resembles 2022, the seasonal-naive baseline will be strong and leaderboard shake-ups are possible. The 548-day horizon exceeds one annual cycle, so 2024-H1 point forecasts carry wider intervals; we will report prediction bands in Part 3.

8. Data audit & feature governance

Beyond insight generation, a robust EDA must hand the modelling stage an *auditable* feature contract. Table 2 summarises the main integrity and missingness facts; Table 3 makes the leakage-driven feature blacklist explicit.

Reproducibility. Everything in this report is regenerated by running `uv run python results/v4/eda_v4.py`. All quoted numbers live in `results/v4/metrics_v4.json`; figures are vector PDFs in `results/v4/images/`. A notebook wrapper is at `results/v4/eda_v4.ipynb`. The pipeline is deterministic (no sampling).

Table 2: Data audit. 14 tables, 2012-07-04 to 2022-12-31. All 10 foreign-key relationships checked; orphan rows $= 0$.

Table	Rows	Max %null	Note
sales	3,833	0%	1 row/day, no gaps
orders	646,945	0%	FK → customers OK
order_items	714,669	99.97%	`promo_id_2` sparse
products	2,412	0%	4 categories, 8 segments
customers	121,930	low	`gender`,`age_group` null $\approx 30\%$
promotions	50	80%	`applicable_category` null
shipments	566,067	0%	only shipped orders
returns	39,939	0%	3.1% of revenue refunded
reviews	113,551	0%	~20% of delivered
inventory	monthly	0%	1 row/SKU/month
web_traffic	daily	0%	no `conversion_rate`

Table 3: Leakage-driven feature blacklist (features *not* usable at forecast time without the specified lag). Source: Finding 6 ranking.

Feature	Reason	Min. lag
$\texttt{COGS}_t$	joint target, $\rho=0.97$	forecast sep.
$\texttt{n\_orders}_t$	contemporaneous, $\rho=0.92$	lag ≥ 1
$\texttt{unique\_customers}_t$	contemporaneous, $\rho=0.92$	lag ≥ 1 / 30d
$\texttt{refund}_t$	correlation, not cause	lag ≥ 30
$\texttt{inventory}_t$ month	snapshot = target month	lag ≥ 30
$\texttt{margin}_t$	derived from target	lag ≥ 90 roll
same-day rolling of R	direct leak	strict past-only